MATHS MANTHAN

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Subject: Mathematics | **WEEKLY CHAPTER TEST – 04 (SOLUTIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. In trapezium $ABCD$, $AB \parallel DC$ and diagonals intersect at O. If $AO = (3x-1)$, $OC = (5x-3)$, $BO = (2x+1)$, and $OD = (6x-5)$, the value of x is: [1]

(a) $\frac{1}{2}$ (b) 2 (c) 3 (d) 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 2

Given: In trapezium $ABCD$ ($AB \parallel DC$), $AO = 3x-1$, $OC = 5x-3$, $BO = 2x+1$, $OD = 6x-5$. We know that diagonals of a trapezium divide each other proportionally (since $\triangle AOB \sim \triangle COD$ by AA similarity):

Formula / Theorem Used: $\frac{AO}{OC} = \frac{BO}{OD}$

Substituting the given expressions and cross-multiplying:

$$\frac{3x-1}{5x-3} = \frac{2x+1}{6x-5} \implies (3x-1)(6x-5) = (2x+1)(5x-3)$$

Expanding both sides step by step:

$$18x^2 - 15x - 6x + 5 = 10x^2 - 6x + 5x - 3 \implies 18x^2 - 21x + 5 = 10x^2 - x - 3$$

Transposing all terms to the LHS and dividing by 4:

$$8x^2 - 20x + 8 = 0 \implies 2x^2 - 5x + 2 = 0 \implies (2x-1)(x-2) = 0 \implies x = \frac{1}{2} \text{ or } x = 2$$

Domain Check: At $x = \frac{1}{2}$, $OC = 5(\frac{1}{2}) - 3 = -0.5 < 0$ and $OD = 6(\frac{1}{2}) - 5 = -2 < 0$. Since segment lengths cannot be negative, $x = \frac{1}{2}$ is rejected. At $x = 2$, $AO = 5$, $OC = 7$, $BO = 5$, $OD = 7$ (all positive). Hence, $\boxed{x = \mathbf{2}}$.

Q2. In $\triangle ABC$, D and E are points on sides AB and AC such that $AD = 2$ cm, $BD = 3$ cm, $AE = 3.6$ cm, and $AC = 9$ cm. Then: [1]

(a) DE is not parallel to BC (b) $DE \parallel BC$ (c) $DE \perp BC$ (d) $\angle ADE = \angle B$ is false

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) $DE \parallel BC$

Given: $AD = 2$ cm, $BD = 3$ cm, $AE = 3.6$ cm, $AC = 9$ cm $\implies$ $EC = AC - AE = 9 - 3.6 = 5.4$ cm.

We know the Converse of Basic Proportionality Theorem:

Formula / Theorem Used: If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.

Computing both ratios:

$$\frac{AD}{DB} = \frac{2}{3} \quad \text{and} \quad \frac{AE}{EC} = \frac{3.6}{5.4} = \frac{36}{54} = \frac{2 \times 18}{3 \times 18} = \frac{2}{3}$$

Since $\frac{AD}{DB} = \frac{AE}{EC} = \frac{2}{3}$, by Converse of BPT, $\boxed{\textbf{DE} \parallel \textbf{BC}}$.

Q3. If $\triangle ABC \sim \triangle QRP$, $\frac{AB}{QR} = \frac{1}{2}$, $BC = 2.5$ cm, and $CA = 3$ cm ($AB = 3$ cm), the perimeter of $\triangle PQR$ is: [1]

(a) 15 cm (b) 17 cm (c) 16 cm (d) 20 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 17 cm

Given: $\triangle ABC \sim \triangle QRP$, $\frac{AB}{QR} = \frac{1}{2}$, $AB = 3$ cm, $BC = 2.5$ cm, $CA = 3$ cm.

We know the perimeter property of similar triangles:

Formula / Theorem Used: $\frac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle QRP)} = \frac{AB}{QR}$

Perimeter of $\triangle ABC = AB + BC + CA = 3 + 2.5 + 3 = 8.5$ cm. Substituting into the ratio:

$$\frac{8.5}{\text{Perimeter}(\triangle PQR)} = \frac{1}{2} \implies \text{Perimeter}(\triangle PQR) = 2 \times 8.5 = \boxed{\textbf{17 cm}}$$

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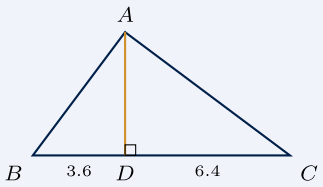
Q4. In $\triangle ABC$, $\angle A = 90°$ and $AD \perp BC$. If $BD = 3.6$ cm and $CD = 6.4$ cm, then the length of altitude AD is: [1]

(a) 4.8 cm (b) 5.0 cm (c) 4.2 cm (d) 5.4 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 4.8 cm

Given: $\angle BAC = 90°$, $AD \perp BC$, $BD = 3.6$ cm, $CD = 6.4$ cm.



Since $AD \perp BC$ in right $\triangle ABC$, $\triangle ABD \sim \triangle CAD$ (AA similarity):

Formula / Theorem Used: $\frac{BD}{AD} = \frac{AD}{CD} \implies AD^2 = BD \times CD$

Substituting $BD = 3.6$ and $CD = 6.4$:

$$AD^2 = 3.6 \times 6.4 = \frac{36}{10} \times \frac{64}{10} = \frac{2304}{100}$$

Taking positive square root:

$$AD = \frac{6 \times 8}{10} = \frac{48}{10} = \boxed{\mathbf{4.8}\text{ cm}}$$

Q5. **Assertion (A):** In $\triangle ABC$, D and E are on AB, AC such that $AD = 3$, $DB = 4.5$, $AE = 2$, $EC = 3$. Then $DE \parallel BC$.

Reason (R): If a line divides two sides of a triangle in the same ratio, it is parallel to the third side. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

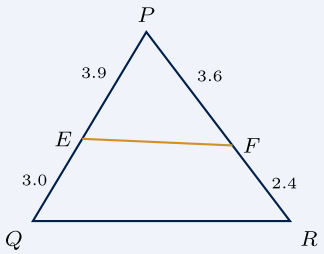
We know the Converse of Basic Proportionality Theorem (stated in Reason (R)):

Formula / Theorem Used: If $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.

Computing the ratios: $\frac{AD}{DB} = \frac{3}{4.5} = \frac{30}{45} = \frac{2}{3}$ and $\frac{AE}{EC} = \frac{2}{3}$. Since $\frac{AD}{DB} = \frac{AE}{EC}$, $DE \parallel BC$. Thus, both (A) and (R) are true and (R) correctly explains (A).

Q6. E and F are points on sides PQ and PR respectively of $\triangle PQR$. If $PE = 3.9$ cm, $EQ = 3.0$ cm, $PF = 3.6$ cm, and $FR = 2.4$ cm, determine whether $EF \parallel QR$, stating the mathematical criterion used. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Declaring the Criterion [0.5 Mark]

Given: In $\triangle PQR$, $PE = 3.9$ cm, $EQ = 3.0$ cm, $PF = 3.6$ cm, $FR = 2.4$ cm.

We know the Converse of Basic Proportionality Theorem:

Formula / Theorem Used: Line $EF \parallel QR$ if and only if $\dfrac{PE}{EQ} = \dfrac{PF}{FR}$.

Step 2: Computing Both Ratios [1 Mark]

$$\frac{PE}{EQ} = \frac{3.9}{3.0} = \frac{39}{30} = \frac{13}{10} = 1.3$$

$$\frac{PF}{FR} = \frac{3.6}{2.4} = \frac{36}{24} = \frac{3}{2} = 1.5$$

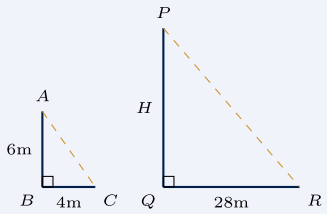
Step 3: Comparison & Conclusion [0.5 Mark]

Since $\dfrac{PE}{EQ} \neq \dfrac{PF}{FR}$ $(1.3 \neq 1.5)$, by Converse of BPT:

$\therefore$ **EF** is NOT parallel to **QR**.

Q7. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Diagram Setup & Similarity Proof [1.5 Marks]

Given: Pole $AB = 6$ m, pole shadow $BC = 4$ m, tower shadow $QR = 28$ m. Let height of tower be $PQ = H$ m.

In $\triangle ABC$ and $\triangle PQR$:

1. $\angle B = \angle Q = 90°$ (Vertical structures on level ground)
2. $\angle C = \angle R$ (Equal sun's elevation at the same time)

By AA similarity, $\triangle ABC \sim \triangle PQR$.

Formula / Theorem Used: $\dfrac{PQ}{AB} = \dfrac{QR}{BC}$

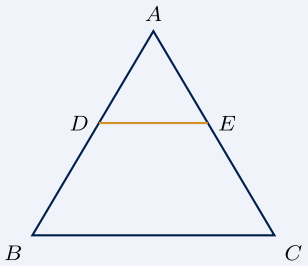
Step 2: Substitution & Calculation [1.5 Marks]

$$\frac{H}{6} = \frac{28}{4} = 7 \implies H = 7 \times 6 = 42 \text{ m}$$

Height of the tower $=$ **42** m

Q8. In the given figure, $\angle D = \angle E$ and $\dfrac{AD}{DB} = \dfrac{AE}{EC}$. Prove mathematically that $\triangle BAC$ is an isosceles triangle. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: In $\triangle BAC$, D lies on AB and E lies on AC such that $\dfrac{AD}{DB} = \dfrac{AE}{EC}$ and $\angle ADE = \angle AED$ ($\angle D = \angle E$).
To Prove: $\triangle BAC$ is an isosceles triangle ($AB = AC$).
We know the Converse of Basic Proportionality Theorem:

> **Formula / Theorem Used:** If $\dfrac{AD}{DB} = \dfrac{AE}{EC}$ in $\triangle ABC$, then $DE \parallel BC$.

Step 1: Establishing $DE \parallel BC$ & Angle Equality [1.5 Marks]
Since $\dfrac{AD}{DB} = \dfrac{AE}{EC}$, by the Converse of BPT, $DE \parallel BC$.
Because $DE \parallel BC$, corresponding angles are equal:

$$\angle ADE = \angle B \quad \text{and} \quad \angle AED = \angle C$$

Step 2: Deducing Equal Sides [1.5 Marks]
It is given that $\angle D = \angle E$, i.e., $\angle ADE = \angle AED$. Substituting from above:

$$\angle B = \angle C$$

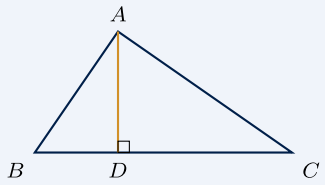
Since sides opposite equal angles in $\triangle BAC$ are equal, $AB = AC$.

$\therefore \triangle BAC$ is an isosceles triangle. [Hence Proved]

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Q9. In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \times CD$. Prove mathematically that $\triangle ABC$ is a right-angled triangle at A. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: In $\triangle ABC$, $AD \perp BC$ ($\angle ADB = \angle CDA = 90°$) and $AD^2 = BD \times CD$.
To Prove: $\triangle ABC$ is right-angled at A, i.e., $\angle BAC = 90°$.

We know the SAS similarity criterion for triangles:

Formula / Theorem Used: If two pairs of sides are proportional and their included angles are equal, the two triangles are similar.

Step 1: Proving $\triangle ABD \sim \triangle CAD$ by SAS [1.5 Marks]

From $AD^2 = BD \times CD$, rewriting as a ratio ($AD \cdot AD = BD \cdot CD$):

$$\frac{BD}{AD} = \frac{AD}{CD}$$

In $\triangle ABD$ and $\triangle CAD$:

1. $\frac{BD}{AD} = \frac{AD}{CD}$ (Proved above)
2. $\angle ADB = \angle CDA = 90°$ (Given $AD \perp BC$)

By SAS similarity, $\triangle ABD \sim \triangle CAD$.

Step 2: Using Angle Sum Property in $\triangle ABC$ [1.5 Marks]

From similarity, corresponding angles are equal:

$$\angle BAD = \angle ACD = \angle C \quad \text{and} \quad \angle B = \angle CAD$$

In $\triangle ABC$, by the angle sum property:

$$\angle BAC + \angle B + \angle C = 180°$$

Substituting $\angle B = \angle CAD$ and $\angle C = \angle BAD$:

$$\angle BAC + (\angle CAD + \angle BAD) = 180° \implies \angle BAC + \angle BAC = 180°$$

$$2\angle BAC = 180° \implies \boxed{\therefore \angle BAC = 90° \quad \text{[Hence Proved]}}$$

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Q10. Case Study 04: Overhead Optical Projector & Quadrilateral Enlargement

An overhead projector bulb is fixed at point O above a presentation table. A transparent slide in the form of quadrilateral $ABCD$ is held horizontally between the bulb and the table top. The projected image forms an enlarged quadrilateral $A'B'C'D'$ on the screen with a scale factor of $1 : 2.5$. The dimensions of the slide are $AB = 2.0$ cm, $BC = 3.0$ cm, $CD = 2.4$ cm, $DA = 1.6$ cm, and $\angle A = 105°$. [4]

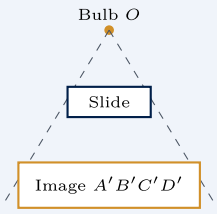
(i) **[1 Mark]** What is the exact measure of $\angle A'$ in the projected image?

(ii) **[1 Mark]** Find the length of side $B'C'$ of the projected image.

(iii) **[2 Marks]** Calculate the total perimeter of the projected quadrilateral $A'B'C'D'$.

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Data: Slide $ABCD \sim$ Image $A'B'C'D'$ with enlargement scale factor $1 : 2.5$ (so each image length is 2.5 times the corresponding slide length).
Slide sides: $AB = 2.0$ cm, $BC = 3.0$ cm, $CD = 2.4$ cm, $DA = 1.6$ cm, and $\angle A = 105°$.
We know the fundamental properties of similar polygons:



Formula / Theorem Used: (1) Corresponding angles are equal ($\angle A' = \angle A$). (2) Each projected side = 2.5 × slide side.

Sub-part (i): Exact Measure of $\angle A'$ [1 Mark]

Since optical projection creates a geometrically similar quadrilateral ($A'B'C'D' \sim ABCD$), corresponding angles remain unchanged:

$$\angle A' = \angle A = 105° \implies \boxed{\angle A' = \mathbf{105°}}$$

Sub-part (ii): Length of Side $B'C'$ in the Projected Image [1 Mark]

Given scale ratio $\dfrac{\text{Slide side}}{\text{Image side}} = \dfrac{1}{2.5}$:

$$\frac{BC}{B'C'} = \frac{1}{2.5} \implies B'C' = 2.5 \times BC = 2.5 \times 3.0 = 7.5 \text{ cm} \implies \boxed{B'C' = \mathbf{7.5} \text{ cm}}$$

Sub-part (iii): Total Perimeter of Projected Quadrilateral $A'B'C'D'$ [2 Marks]

Computing all four enlarged sides using the scale factor 2.5:

$A'B' = 2.5 \times 2.0 = 5.0$ cm, $B'C' = 7.5$ cm, $C'D' = 2.5 \times 2.4 = 6.0$ cm, $D'A' = 2.5 \times 1.6 = 4.0$ cm

Summing all four sides (or checking $2.5 \times \text{Perimeter}(ABCD) = 2.5 \times 9.0 = 22.5$ cm):

$$\text{Perimeter}(A'B'C'D') = 5.0 + 7.5 + 6.0 + 4.0 = 22.5 \text{ cm} \implies \boxed{\text{Perimeter}(A'B'C'D') = \mathbf{22.5} \text{ cm}}$$

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